Lesson notes · for you

# IF this, THEN that

IoT · Level 3 · Lesson 1

**Materials**
Paper and a pen; A ruler, to draw the day's timeline; Optional: a phone or computer for the Smart classroom rules lab

## Learning goals

- Write a rule as IF a condition THEN an action.
- Combine conditions with AND and OR to make better rules.
- Test rules through a day and fix the ones that go wrong.

## Activity: Rules for a day

1. Choose a device you can watch at home or at school: a ceiling fan, a security light, a water-tank pump or a garden tap. Write in one sentence what it should do.
2. Write three rules as IF a condition THEN an action. Each condition needs something a sensor can measure, with a number and a unit: “above 30 °C”, “below 20%”, “after 18:00”.
3. Make one rule smarter with AND or OR. Ask: must both things be true (AND), or is either one enough (OR)?
4. Draw a line across your page and mark the hours from 6:00 to 22:00. At each hour, write what each rule makes the device do. Don't forget the breaks, the evening and days when nobody is there.
5. Circle every hour where the device does the wrong thing. Change the rule (a new condition, AND or OR, a different number) and test the whole day again.

**Paper is enough for this activity. Watch real devices only from a safe distance: never open a fan, a pump or a socket, and never touch mains wiring.**

## Discussion

- Which rule around you is too simple, and what would you add to it?
- When is OR safer than AND, and when is AND safer than OR?
- What should a device do when its sensor breaks?

## Check yourself

- Write one rule that uses AND, and say when it switches on.

**Try it this week**
Rules for a day: Choose a device at home or at school: a fan, a light, a water pump or a garden tap. Write three IF/THEN rules for it, with a number and a unit in each, and use AND or OR at least once. Draw a timeline of one day from 6:00 to 22:00, play each rule through it hour by hour, mark where a rule does the wrong thing, and fix it.

Worksheet

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Name _____ Class _____ Date _____

**1** **A rule says: IF the room is above 30 °C, THEN open the window. On a test day, it opens in a rainstorm and soaks the desks. What's the best fix?**

◯ Add: OR it isn't raining ◯ Add: AND it isn't raining ◯ Raise the threshold to 35 °C

**2** **Rule: IF it is dark AND a class is in, THEN lights on. At 6:30 the room is dark and empty, so the lights stay off.**

◯ True ◯ False

**3** **Which of these could a device check with a sensor, and use in a rule?**

☐ The soil moisture is below 30% ☐ The plants look happy ☐ The water tank is below 20%

☐ The class is bored ☐ Someone is moving near the door

**4** **Complete the two rules: the fan runs only for a class, and the alarm rings for either sign.**

IF above 30 °C _____ a class is in, THEN fan on. IF smoke is detected _____ the fire button is pressed, THEN ring the alarm.

**Word bank:** OR · THEN · AND · NOT

**5** **Put the steps for testing a rule in order.**

☐ List times through the day to test this rule, including early morning and evening

☐ Write the rule with a number and a unit

☐ Mark every time it does the wrong thing

☐ Fix the rule, then test the whole day again

☐ At each time, work out what the device would do

✎ **Which device at home or at school would you most like to make automatic? Write its rule, and one time of day when it might go wrong.**

Check your answers

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**1** **Add: AND it isn't raining**

With AND, the window opens only when it is hot and dry.

**2** **True**

AND needs both conditions. It is dark, but no class is in, so the lights stay off and the school saves electricity.

**3** **The soil moisture is below 30%; Someone is moving near the door; The water tank is below 20%**

Moisture, movement and water level can all be measured, so they make good conditions.

**4** **IF above 30 °C AND a class is in, THEN fan on. IF smoke is detected OR the fire button is pressed, THEN ring the alarm.**

The fan needs both, heat and a class, so AND.

**5** **Top to bottom: 2, 1, 4, 5, 3**

You can't test a rule until it is exact, and you can't judge it until you've played it through the day.